

Project Proposal: Bayesian Topic Modeling and Uncertainty Quantification across MS MARCO and BEIR

1. Dataset Selected

This project will utilize two primary datasets commonly used in Information Retrieval (IR) research, which can be accessed via online repositories:

- **MS MARCO (MicroSoft MAchine Reading COmprehension):** A randomly sampled subset of the MS MARCO dataset will serve as the primary training and analysis set. This dataset contains real-world web search queries and document snippets.
- **BEIR (Benchmarking Information Retrieval):** A heterogeneous benchmark consisting of diverse retrieval tasks (e.g., SciFact for science, FiQA for finance). This will be used to evaluate model performance and uncertainty on out-of-domain data.

2. Project Goals and Questions

The goal of this project is to apply hierarchical Bayesian modeling to understand thematic structures in text and quantify how model "confidence" changes across different data domains.

- **Question 1:** What is the latent topic distribution of a random subset of the MS MARCO dataset when analyzed using a Bayesian Latent Dirichlet Allocation (LDA) model?
- **Question 2:** When the LDA model trained on MS MARCO is applied to specialized datasets within the BEIR benchmark, do the posterior credible intervals for topic proportions significantly widen, indicating increased model uncertainty? We will quantify this uncertainty using both the width of 95% posterior credible intervals and Shannon Entropy of the topic distributions. If so, which BEIR benchmark has the most significant effect on the model?

3. Tentative Plan of Analysis

- **Model Selection:** We will implement a **Latent Dirichlet Allocation (LDA)** model, which is a generative hierarchical Bayesian model.
- **Priors:** I will use Dirichlet priors (α and β) for the document-topic and topic-word distributions. We plan to experiment with both symmetric and asymmetric priors to see which better captures the sparsity of the data.
- **MCMC Sampling:** We will use **Stan (R)** or **JAGS (R)** to perform **Markov Chain Monte Carlo (MCMC)** using state of the art MCMC algorithms to estimate the posterior distributions of the latent variables. We will monitor convergence using trace plots and the Gelman-Rubin statistic ($\hat{R}$) to ensure the chains have adequately explored the posterior space.
- **Model Validation:** We will perform Posterior Predictive Checks to see if the generated data matches the observed word frequencies in the MS MARCO subset.
- **Prediction/Comparison:** The trained model will be used to infer topic proportions for documents in the BEIR benchmark. We will calculate the variance of the posterior distributions to compare the width of the 95% credible intervals between in-domain (MS MARCO) and out-of-domain (BEIR) data.

4. Expected Outcomes

- **MS MARCO Distribution:** We expect the analysis of MS MARCO to reveal a distribution of topics centered around common web search intents, such as "informational," "transactional," and "navigational" clusters.
- **Uncertainty Quantification:** We anticipate that the LDA model will exhibit significantly wider credible intervals (increased posterior variance) when applied to the BEIR datasets. Because BEIR contains specialized terminology (e.g., medical or legal) not present in general web searches, the model should express higher "epistemic uncertainty" regarding topic assignments for those documents.